Customer Churn Analysis

Churn rate, Customer Lifetime Value, revenue impact and behavioural drivers

Customers	5,000	**Churn rate**	27.5%
Churned customers	1,376	**Monthly churn rate**	0.97%
ARPU (monthly)	$54.84	**Avg customer lifespan**	103 months
Customer Lifetime Value	$5,674	**MRR lost to churn**	$85,831 (31.3%)

Executive summary

• Overall churn is 27.5% (1,376 of 5,000 customers), costing $85,831 in monthly recurring revenue (31.3% of MRR, about $1,029,970 annualised).

• Customers stay on average 103 months at an ARPU of $54.84, giving a Customer Lifetime Value of $5,674.

• Churned customers left after 20.4 months on average versus 31.5 months of tenure for retained customers.

• 'Month-to-month' contracts churn at 41.5% versus 5.0% for 'Two year' - a 8.3x gap; they account for 84.2% of all churners.

Top recommendations

1. **Move customers off short-term contracts**
2. **Fix the first-year experience**
3. **Promote automatic payments**
4. **Review pricing and bundle value**

1. Data & methodology

The dataset was cleaned and validated before analysis:

• Loaded 5,025 rows x 17 columns; headers standardized to snake_case.

• Trimmed whitespace and converted blank/placeholder strings (e.g. ' ', 'N/A') to nulls.

• Removed 25 duplicate records.

• Encoded churn target as 0/1 (0 rows with unreadable target dropped).

• Imputed 12 missing monthly charges with the median.

• Repaired 51 missing total charges as tenure x monthly charges (brand-new customers = 0).

• Final clean dataset: 5,000 rows (from 5,025); no remaining nulls in analysis fields.

Metric definitions

• **Churn rate** = churned customers / total customers.

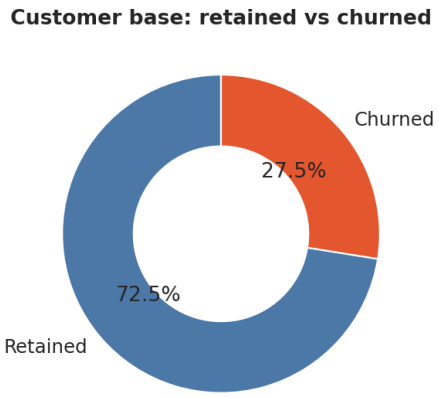
• **Monthly churn rate** = churn events / total customer-months observed (sum of tenure).

• **Average lifespan** = 1 / monthly churn rate.

• **CLV** = ARPU x average lifespan (revenue basis; no margin applied).

• **MRR loss** = sum of monthly charges of churned customers.

2. Baseline churn and revenue impact

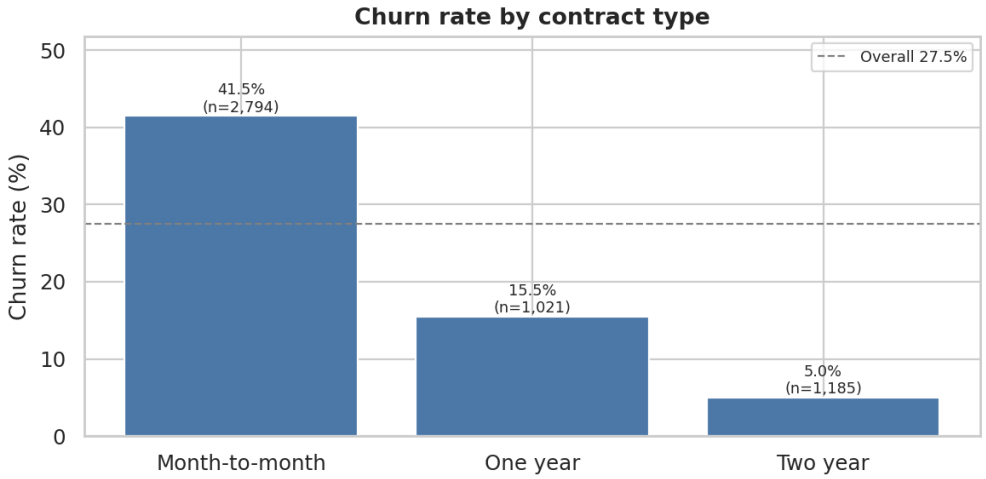


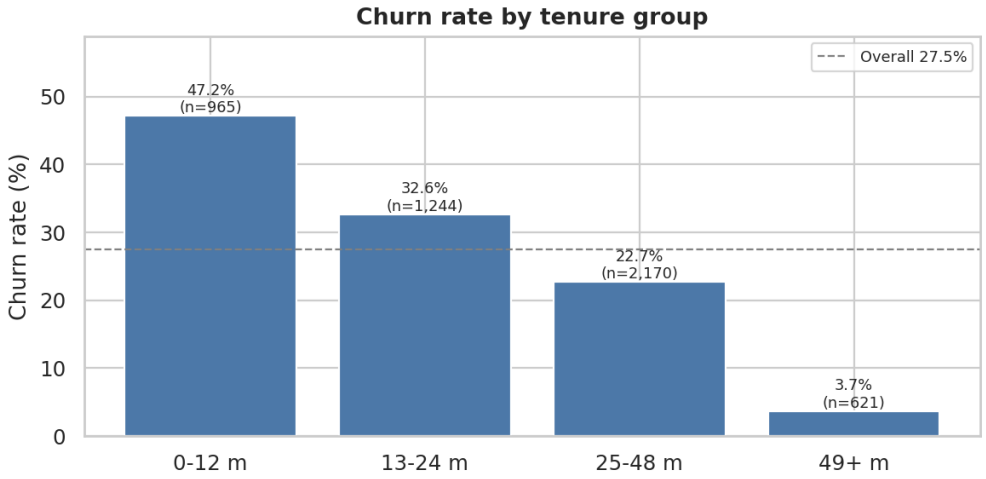
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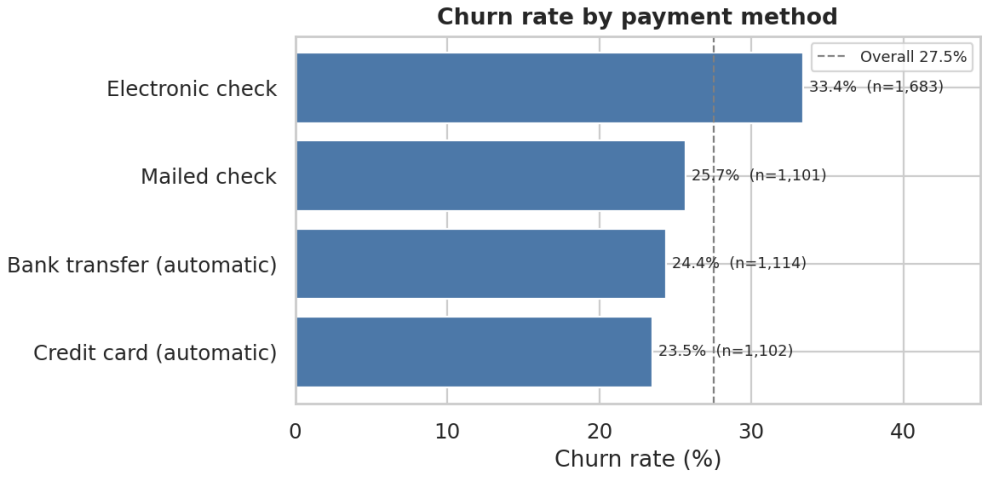
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3. Churn by contract, tenure and payment method





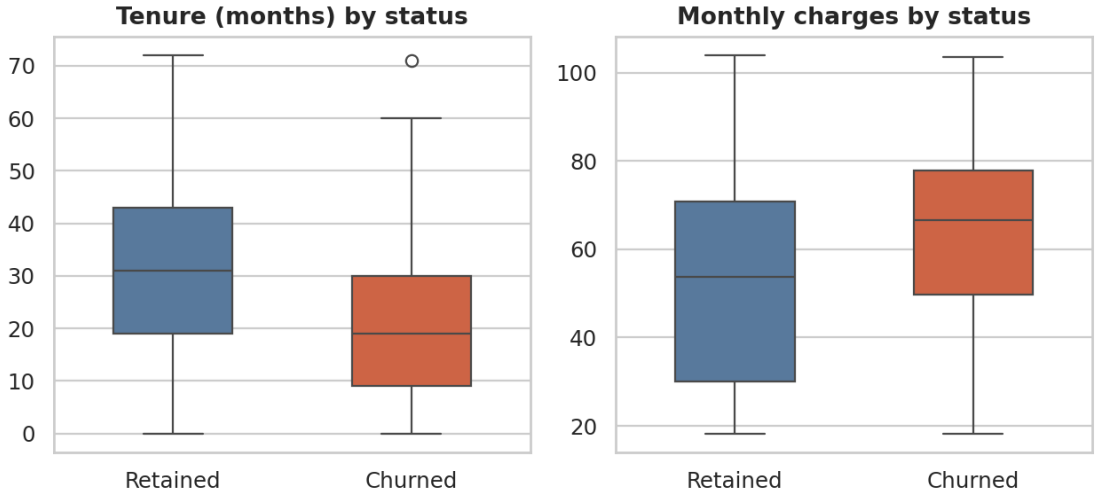


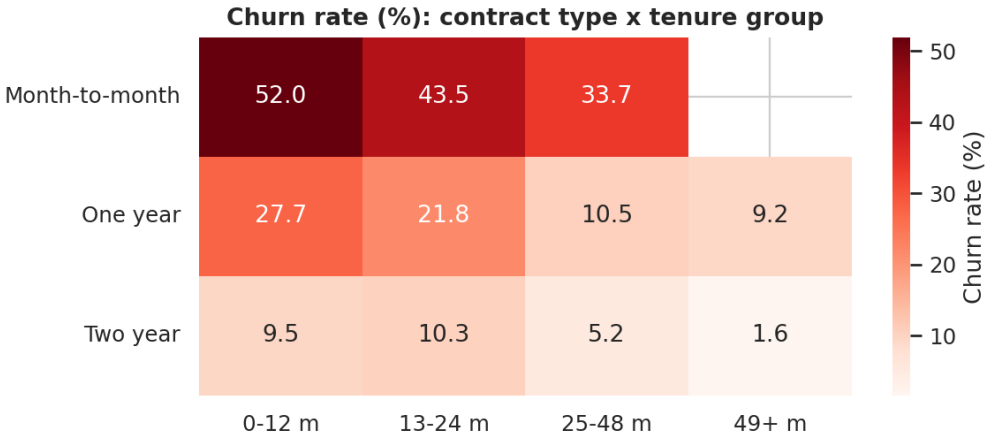
• 'Month-to-month' contracts churn at 41.5% versus 5.0% for 'Two year' - a 8.3x gap; they account for 84.2% of all churners.

• Churn is front-loaded: the '0-12 m' group churns at 47.2% and holds 33.1% of all churners; risk falls steadily as tenure grows.

• 'Electronic check' users churn at 33.4% versus 23.5% for 'Credit card (automatic)'.

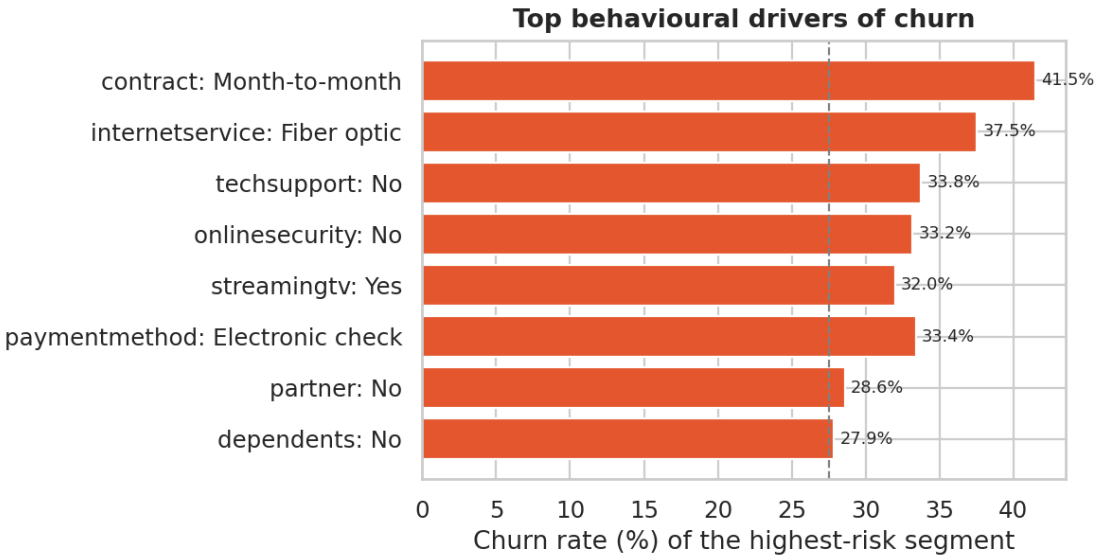
4. Behavioural patterns

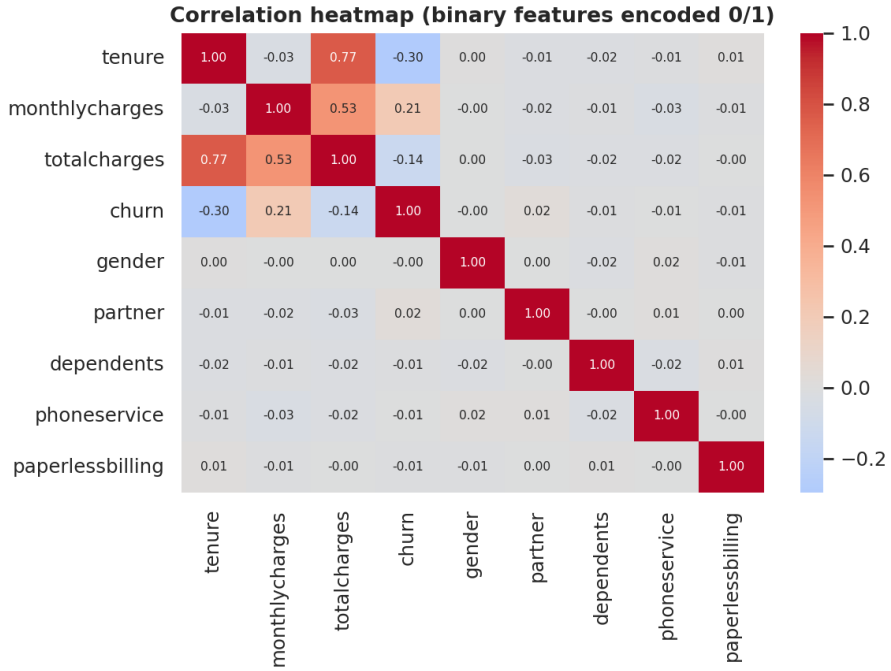




Short-tenure, month-to-month customers form the highest-risk pocket; churn drops sharply with longer contracts and longer tenure.

5. Key drivers of churn





Tenure and contract length show the strongest negative relationship with churn; short-contract and high-charge indicators are positively related.

6. Actionable recommendations

1. Move customers off short-term contracts

'Month-to-month' customers drive 84.2% of churn. Offer 10-15% discounts or a free add-on for 12/24-month commitments. Converting just 10% of the churn in this segment protects roughly $7,155 MRR ($85,862 per year).

2. Fix the first-year experience

33.1% of churners leave in the first 12 months. Introduce a 90-day onboarding programme: welcome call, setup help, a day-30 and day-60 health check, and proactive outreach for low-usage accounts.

3. Promote automatic payments

Customers paying by 'Electronic check' churn 1.4x more than 'Credit card (automatic)' customers. Offer a small bill credit for switching to automatic payment, which reduces friction and failed-payment churn.

4. Review pricing and bundle value

Churners pay more per month ($66.56 vs $53.81). Introduce loyalty pricing, bundles, or a right-sizing review for high-bill accounts before they cancel.

5. Target the high-risk 'internetservice' segment

Customers with internetservice = 'Fiber optic' churn at 37.5% (vs 13.8%). Bundle or trial the related service/benefit for this group and track the change in churn.

6. Build an early-warning model and retention playbook

Score customers monthly using tenure, contract, payment method and charges. Route the top-risk decile to a retention team with pre-approved offers, and measure results with an A/B test against a holdout group.

Next steps: pilot the top two initiatives on a test group, track churn monthly against a control group, and refresh this report quarterly.

Appendix: segment tables

Tenure Group

Tenure Group	Customers	Churned	Churn rate
0-12 m	965	455	47.2%
13-24 m	1,244	406	32.6%
25-48 m	2,170	492	22.7%
49+ m	621	23	3.7%

Contract

Contract	Customers	Churned	Churn rate
Month-to-month	2,794	1,159	41.5%
One year	1,021	158	15.5%
Two year	1,185	59	5.0%

Paymentmethod

Paymentmethod	Customers	Churned	Churn rate
Electronic check	1,683	562	33.4%
Mailed check	1,101	283	25.7%
Bank transfer (automatic)	1,114	272	24.4%
Credit card (automatic)	1,102	259	23.5%